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(1) Research Program/Project/Study Title: Effectiveness of Squash as feed additive in a Fishball like feed on the growth performance of Mangrove Crab <i>Scylla serrata</i> Fattening in a Land-based Vertical Culture System	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	
Province:	
Region:	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)

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(7) Sector/Commodity Fisheries/Mangrove crab	(8) Discipline/Field of Specialization
(9) Rationale The Mangrove crab <i>Scylla serrate</i> is one of the most popular and expensive seafood in Southeast Asian Countries (Sarower <i>et al.</i>, 2013). As an export commodity, Mangrove crab has a relatively high selling price in domestic and foreign markets. However, it all depends on the quality or the size of the crab's fatness. Therefore, the fattening cultivation business is up-and-coming. (Aslamya <i>et al.</i>, 2021). The culturing or fattening of mud crabs (<i>Scylla serrata</i>) is an emerging industry (Azam <i>et al.</i>, 1998; Khan <i>et al.</i>, 1991). The mud crab is widely distributed in the Indo-West-Pacific region from Hawaii, Southern Japan, Taiwan, and the Philippines to Australia, the Red Sea, and East and South Africa (Motoh, 1979; Aiyun & Silliang, 1991; Macintosh <i>et al.</i>, 2002). However, only the mud crab inhabiting marine or brackish waters is commercially important. Mud crabs occur abundantly in the whole coastal region of Bangladesh, particularly in the estuaries and tidal rivers of the Sundarbans shrimp polders (Khan & Alam, 1991). Large-clawed mud crabs are high-commodity seafood items due to their delicacy, medicinal value, and demand in international markets (Ali <i>et al.</i>, 2004; Keenan <i>et al.</i>, 1997). The mud crab aquaculture industry in the province of Capiz received much attention after production constraints due to diseases encountered by the prawn industry. The techniques for grow-out culture were gradually refined as farmers learned from the experiences of others. Presently, mud crab aquaculture is being done by private sectors based more on art. However, mud crab culture requires much science to make the production consistent and sustainable. The practices of farmers and the problems they have encountered are presented together with the overall industry concerns (Hocson, 2017). Mangrove crab fattening is a relatively new practice in the Philippines. Mangrove crab fattening pertains to culturing Mangrove crabs from 15 days to 1 month so that they put on additional weight after molting (Gabito <i>et al.</i>, n.d.). <i>Scylla</i> species have been farmed in ponds, cages, and Pens, but production is erratic (Fortes, 1999). It has been established that rearing crabs in cages and other containers is both technically and financially possible. Land-based crab fattening project may obtain an average growth increase of 20-25 g per crab in 14-15 days with the average stock weighing 200 grams (Morales & Barbara Jr., 2015) As fishers have done so far, the culture of crabs in ponds shows some disadvantages, such as requiring large areas, exposure to pollution from open waters, high penetration of sunlight to crabs, cannibalism, and crabs escaping from ponds still high, burrowing habit causing difficult harvest, low security, and wasteful energy (Datta <i>et al.</i>, 1999). The improvement or innovation of the crab culture system through the crab farming technology in battery boxes immersed in the pond still shows some weaknesses, such as high mortality rate, decreased water quality due to significant remnant food, and exposed to pollution from open waters, crab exposed by direct sunlight resulting in high mortality, flawed water recirculation system, low land efficiency, low-security level, and low work comfort	

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(Sunaryo *et al.*, 2007; Begum *et al.*, 2009). Then, it is necessary to introduce an advanced system in crab culture by operating the flow or recirculation of land-based cultivation facilities (Shelley & Lovatelli, 2011). The crab bucket recirculating water aquaculture system is an advanced system that provides individual aquaculture cages for each crab to avoid the crabs between mutual killings.

Finding low-cost feed supplies has been identified as one of the critical challenges facing the rapid growth of the Mangrove crab farming sector in Asian Countries (Le Vay *et al.*, 2007; Capiña, 2018). The available commercial feeds are primarily based on fishmeal, which is becoming scarce day by day, resulting in a price hike in crab feed (Kar *et al.*, 2017). Hence, searching for an alternative feed for crab feed will solve this problem.

Squash (*C. maxima*) Seed, Bark, and pulp protein content are 7.93 ± 0.02 , 12.63 ± 0.04 and $13.42 \pm 0.02\%$ obtained as protein contents of seed, bark, and pulp, respectively; these values were lower than the 25.89% reported for reported for Cucurbita mixta by Shobha, 2012. However, the values were higher than the $3.89 \pm 0.58\%$ reported for healthy Cucumis melo var. agrestis by Adekunle and Oluwo, 2008. These values were lower than values (14-42%) for seeds and (0.2-2.7%) for the fruit pulp, which is higher as reported for C maxima by Karanja *et al.*, 2014.

The aim of this study is to assess the potential of sustainable Mangrove crab fattening by comparing the growth rate and weight gain of Mangrove crab feed with low-cost artificial diet formulated from squash and trash fish as main protein source in a land-based Mangrove crab culture system.

(9-A) Project Description

This study will formulate a fishball like feed from squash and trash fish for Land-based Mangrove crab fattening, also known as, vertical Mangrove crab fattening method,

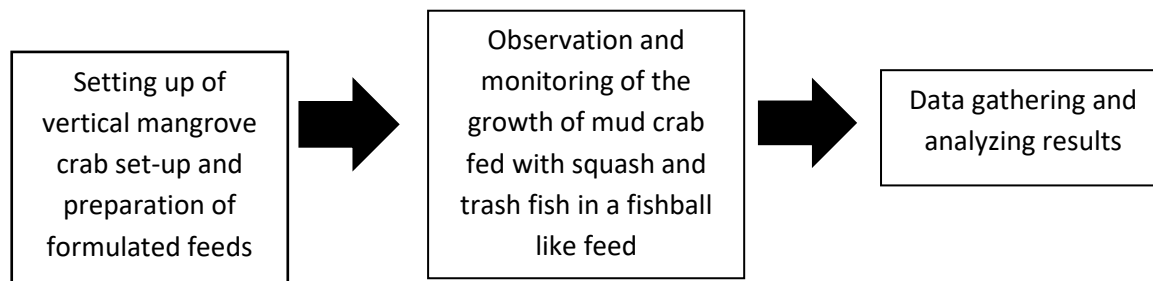
(10) Objectives/Statement of the Problem

Generally, this study is focused on the utilization of squash and trash fish in a fishball like feed and its effects on growth and proximate composition of Mangrove Crab *Scylla serrata* fattened in a land-based Mangrove Crab fattening culture system. Specifically, this study aims to:

1. Determine the efficiency of fishball like feed as protein source for artificial feed for vertical Mangrove crab fattening,
2. Determine the difference in the growth (carapace length and width) and weight of the Mangrove crab fed with trash fish, tadpole, and earthworm formulated feeds,
3. Determine if there is a significant difference in the growth (carapace length and width) and weight of the Mangrove crab using trash fish, tadpole, and earthworm-formulated feeds,
4. Determine the Proximate Biochemical Composition (Moisture, Ash, Protein, Fat, and Carbohydrate) of the fattened Mangrove crab.

(11) Scientific Basis/Theoretical Framework

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(13) Methodology

1. Feed formulation and culture period will be carried out in the FFRDC laboratory. Proximate biochemical analysis of the fattened Mangrove crabs will be sent to DOST Laboratories.

2. **Methods.** This study will be arranged in a Randomized Complete Block Design (RCBD) with 5 treatments and 3 replications.

Treatments:

1. Trash fish meat (Control)
2. Trash fish meal-formulated feeds
3. Garden snail meal-formulated feeds
4. Earthworm meal-formulated feeds
5. Garden snail meal and Earthworm meal formulated feeds

3. **Procedure.** The first phase consists of preparing the vertical Mangrove crab set-up and the preparation of feed formulation will be executed. Person's Square Technique will be used in determining the Protein percent and the weight of the ingredients needed in the preparation of the feeds. Dry ingredients will be grinded separately according to each treatment. Grinded dry ingredients will then be mixed with binders and other liquid ingredients. The mixture will then be heated over low heat to obtain a pasty texture. The mixture will be molded to the desired shape and size. Molded feeds will be dry through the process of sun drying, dehydrator, or air drying.

Ingredients:

Treatment 1	Treatment 3	Treatment 4	Treatment 5
Fish meat	Fish meat	Fish meat	Fish meat
Wheat flour	Wheat flour	Wheat flour	Wheat flour
Egg	Egg	Egg	Egg
Water	Water	Water	Water

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4. Feeding and management. Fresh tidal water from creek will be used for water exchange of vertical culture system. Water quality parameters such as Temperature, pH, Salinity and Dissolved Oxygen (DO) will be monitored daily using portable instruments. The boxes will be cleaned daily by removing excess feeds and waste.

5. Sampling and Harvest. Mangrove Crabs will be assessed every 5 days to record individual increases in body weight and growth rate and to make adjustments on feeding. After the fattening period (15 days), all the crabs will be harvested individually by hand. Total WG, percent WG, SGR, FCR and survival rate will be calculated according to Idonibouye-obu *et al.* 2003, SGR according to Houlihan *et al.* 2001 and FCR according to Henken *et al.*, 2007.

6. Data Gathering and Analysis. Formulated feeds, fresh Mangrove crab and steamed Mangrove crab will be sent to DOST Laboratory for the Proximate Biochemical Composition. Data gathered will be analyzed and encoded. ANOVA will be used to observe any differences in growth parameters of crabs.

(14) Major Activities/Work plan

Research Objectives	Weeks										
	1	2	3	4	5	6	7	8	9	10	11
1. Preparation of the vertical Mangrove crab fattening set-up											
2. Procurement and Selection of materials to be used in feed preparation.											
3. Feed Formulation											
4. Selection of Mangrove Crab samples											
5. Culture period											
6. Data gathering, encoding and analysis											
7. Preparation of Manuscript, IEC materials and for patent/copyright, UMs											

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(15) Expected Output

1. Formulated feeds
2. Publication
3. IEC Materials
4. Copyrighted materials

(15) Technology Transfer Activity Design

Conduct training on the process of making feeds and recirculating aquaculture system.

(16) Targets Beneficiaries

Aquaculturists and Mangrove Crab growers in the province of Capiz

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
STEPHANIE S. PIMENTEL	Study Leader	Biology

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (Php)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		5,000.00
- Transportation allowance		5,000.00
Bamboo	30 pcs.	3000.00
Pawod	100 pcs.	1000.00
Basin	120 pcs.	9600.00
PVC 1 inch	20 pcs.	2000.00
PVC 3 inches	2 pcs.	500.00
UV Lights	2 pcs	2000.00
Nails 4 inches	2 kg	1000.00
Nails 1 1/4 inches	1.5 kg	225.00
T- Hose	10 pcs	400.00
Elbow 1"	20 pcs	400.00
Elbow 3"	50 pcs	600.00
Union Pipe	120 pcs	2400.00

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Vulcaseal	2 liter	1000.00
Aerator	2 pcs	2000.00
Submersible Pump	2 pcs	1000.00
K1 Media	30 kg	15000.00
Coupling Pipe	10 pcs	150.00
Brush Filter	40 pcs	2400.00
Pail	8 pcs	2400.00
Aerator Host	8 pcs	160.00
Labor	2 person for 3 days	3000.00
Binder	1 roll	150.00
Net	5 meters	250.00
Professional fees (laboratory analysis and other related fees)	1 pc. Feed & 1 pc. Cooked specimen	20,000.00
Miscellaneous (for incidental expenses)		5,000.00
Patent Application fee		10,000.00
Journal fee		25,000.00
C. CAPITAL OUTLAY		
Total		115,635.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Ali, S. A., Dayal, J. S., & Ambasankar, K. (2011). Presentation and evaluation of formulated feed for Mangrove crab *Scylla serrata*.

Aslamyiah, S., Hidayani, A. A., Azis, H. Y., & Fujaya, Y. (2021, May). The organoleptic, physical and chemical quality of mud crab fattening feed fermented with a microorganism mixture. In *IOP Conference Series: Earth and Environmental Science* (Vol. 763, No. 1, p. 012032). IOP Publishing.

Audu, R., & Yola, I. A. (2021). Contemporary issues in fisheries and aquaculture: a review on non-conventional feed ingredients for fish feed in Nigeria. *Bayero Journal of Pure and Applied Sciences*, 13(2), 22-28.

Capiña, M. M. (2018). Effect of Golden Apple Snail (*Pomacea canaliculata*) Meat as Feeds on the Growth and Survival Rates and Biochemical Composition of Mud Crab (*Scylla serrata*). *Journal of Innovative Research in Clinical & Medical Sciences-ISSN*, 2456, 7736.

Fortes, R. D. (1999). Mud crab research and development in the Philippines: an overview. In *ACIAR PROCEEDINGS* (pp. 27-32). AUSTRALIAN CENTRE FOR INTERNATIONAL AGRICULTURAL.

GABITO, P. P., SENTINA, F. C., & LUNA, N. D. THE GROWTH CAPACITY OF MUD CRAB (*Scylla serrata*) IN CAGES UNDER MANGOVE AND AT OPEN CAMPOCPOC BAY.

Hindatu, A., Solomon, R. J. (2016) The use of tadpole meal as a substitute for fish meal in diets of (*clarias gariepinus*) fingerlings. International Standard Journal Number (ISJN) e-ISJN: A4372-2604.

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Imbuk, E. S., Indanan, S.L., Sailadjan, S.J, Yürüten Özdemir, K., Sarri, J.H. (2023). Fattening of Mangrove Crab *Scylla serrata* Fed with Two Different Diets (Stingray and Trash Fish). *MedFAR.*, 6(1):1-9

Kar, S., Salam, M. A., & Rana, K. S. (2017). Artificial feed development through fishmeal replacement with non-conventional feed stuff for mud crab (*Scylla serrata*) fattening. *International Journal of Applied Research*, 3(6), 237-242.

Morales, M. I., & Barba Jr, R. B. (2015). Effects of photoperiod, water levels and sex on the feeding efficiency and weight increment of mudcrabs (*Scylla serrata* Forskall) in a crab-fattening culture system. *International Journal of Fisheries and Aquatic Studies*, 3(1), 320-324.

Sarower, M. G., Bilkis, S., Rauf, M. A., Khanom, M., & Islam, M. S. (2013). Comparative biochemical composition of natural and fattened mud crab *Scylla serrata*. *Journal of Scientific Research*, 5(3), 545-553.

Sogbesan, A. O., & Ugwumba, A. A. A. (2008). Nutritional values of some non-conventional animal protein feedstuffs used as fishmeal supplement in aquaculture practices in Nigeria. *Turkish Journal of Fisheries and Aquatic Sciences*, 8(1), 159-164.

Yulianto, B., Sunaryo, Nur Taufiq S.P.J., Ali Djunaedi, Subagiyo, Adi Santosa and Agoes Soegianto (2019). Survival and Growth of mud crab (*Scylla serrate* Forsskål, 1775) reared in crab bucket recirculating aquaculture system

(20) Capsule Curriculum Vitae (One-page brief C.V. only)